Building intelligent Global Health and
Humanitarian Response Technologies

Week 6:
Reasoning Against Ourselves

Context **The *Confident Fool*: How Expertise Breeds Error**
Engineering **Building Systems to Make *Reliable* High Stakes Decisions**

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EPFL ARIADNE LABS ASHOKA UNIVERSITY WORLD ECONOMIC FORUM Rwanda Centre for the Fourth Industrial Revolution

The *Confident* Fool: How Expertise Breeds Error



Lecturer: Yusuf Kesmen

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CONTEXT

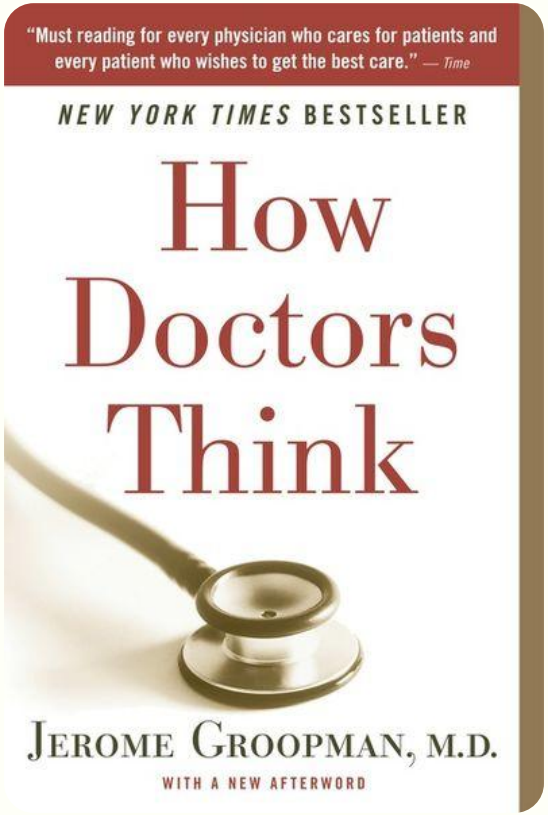
Why was Anne Dodge still ill?

Anne Dodge

- **15 years** of illness.
- Prior labels: **eating disorders**.
- Weight loss despite reported high food intake.

What does the label explain?

What remains unexplained?



Groopman, 2007

CONTEXT

This is not just one unusual case

≈1 in 20

US adults experience an outpatient **diagnostic error each year**, by a 2014 estimate.



Singh et al., 2014

CONTEXT

Autopsies reveal major missed diagnoses

23.5%

Median error rate

53 autopsy series, 1966–2002



Shojania et al., JAMA 2003

CONTEXT

How does a diagnosis take shape?

Before explaining errors, follow the ordinary reasoning process:

Represent	→	Generate	→	Investigate	→	Decide
the patient's problem		possible explanations		ask, compare, update		act now or defer

CONTEXT

First, turn the story into a clinical problem

I twisted my right knee yesterday while playing football.



Bordage, 1994

CONTEXT

First, turn the story into a clinical problem

I twisted my right knee yesterday while playing football.





Acute
Yesterday

Right
Right knee

Traumatic
Sports-related

Twisting
Mechanism

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Bordage, 1994

CONTEXT

Semantic qualifiers

- Organized as **structured knowledge in semantic axis**
- **Abstract representation** of presenting patient state
- Can be lossy

Time Course	Acute	Chronic
Laterality	Unilateral	Bilateral
Cause	Traumatic	Atraumatic

CONTEXT

Semantic qualifiers

- Organized as **structured knowledge in semantic axis**
- **Abstract representation** of presenting patient state
- Can be lossy

Time Course	**Acute**	Chronic
Laterality	**Unilateral**	Bilateral
Cause	**Traumatic**	Atraumatic

CONTEXT

The representation cues disease knowledge

- Doctors store the clinical knowledge in a structured database in the brain
  - **Context:** Who is at risk, and in what setting?
  - **Mechanism:** What process could explain the findings?
  - **Predictions:** What else should we expect?
- Semantic qualifiers activate the scripts for illnesses

Schmidt et al., 1990

CONTEXT

Illness script

twisted, acute, right knee, traumatic, yesterday, football



ACL
Pivoting sports
Ligament Tear
Rapid Swelling

Meniscal
Twisting
Meniscal Tear
Locking

Collateral Ligament
Sideway knee
Ligament Damage
Laxity

CONTEXT

Illness script

twisted, acute, right knee, traumatic, yesterday, football



ACL
Pivoting sports
Ligament Tear
Rapid Swelling

Meniscal
Twisting
Meniscal Tear
Locking

Collateral Ligament
Sideway knee
Ligament Damage
Laxity

CONTEXT

Same patient, different knowledge structures

I twisted my right knee yesterday while playing football.



A twisting injury during football? Did you hear a pop or experience immediate swelling?



Expert:
Illness Scripts

Twisting puts rotational stress on the knee. That could damage the meniscus or ligaments.



Novice:
Biomedical Causal Networks

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CONTEXT

Why are some clinicians better?

- Even if they have the same knowledge:
  - They **structure** the information differently
  - They **reason** on the information differently

CONTEXT

How do clinicians reason?

System 1

Recognition

Fast, effortless, intuitive

“This resembles pneumonia.”

System 2

Comparison

Slow, effortful, analytical

“What supports it?
What conflicts? What else fits?”

This is called as **Dual-Process** theory

Kahneman, 2011 · Norman et al., 2017

CONTEXT

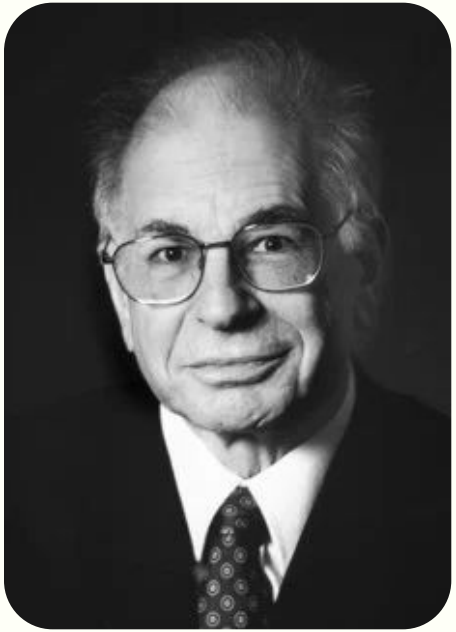
Daniel Kahneman (1934–2024)

1974

Heuristics and biases

2002

Nobel Prize for Economic Sciences



Human use *shortcuts* in reasoning which leads certain *biases*

Tversky & Kahneman, 1974 · Kahneman, Nobel biography, 2002

CONTEXT

Where can judgement fail?

Different failures can interrupt the same diagnostic process.

- **Framing** — Wording changes our judgment.
- **Availability** — Easy recall changes what comes to mind.
- **Anchoring** — The first explanation resists revision.
- **Search satisfaction** — The search stops too soon.
- **Overconfidence** — Confidence exceeds accuracy.
- **(...)**

CONTEXT

Which report feels safer?

REPORT A

90 / 100

patients survive

A This feels safer

REPORT B

10 / 100

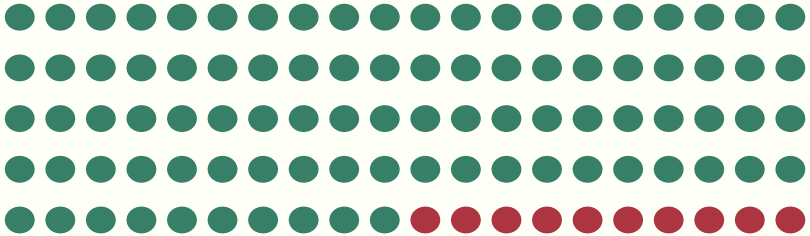
patients die

B This feels safer

CONTEXT

Which report feels safer?

Same Distribution



- 90 survive
- 10 die

Judgement can shift when equivalent information is expressed as a gain or a loss.

CONTEXT

Framing: the numbers did not change

HANDOFF VERSION A

EMERGENCY DEPARTMENT NOTE

A patient with an **anxiety history** presenting with new breathlessness and tachycardia.

HANDOFF VERSION B

EMERGENCY DEPARTMENT NOTE

New **breathlessness and tachycardia** in a patient with an anxiety history.

Tversky & Kahneman, 1981 · Croskerry, 2002

CONTEXT

Quick vote: where does R occur more often?

R

Try to recall words

A **R is the first letter**
Rain, river, road…

B **R is the third letter**
Try recalling these.

CONTEXT

Availability: recall is not probability

B: the third letter is more frequent.

- Words beginning with R are easier to retrieve.
- We mistake **ease of recall** for frequency.

Easy to recall ≠ more likely

Tversky & Kahneman, 1974

CONTEXT

Clinicians tend to remember yesterday

YESTERDAY		TODAY
MEMORABLE CASE		NEW PATIENT
A dramatic pulmonary embolism in a breathless patient.	→	Breathlessness again. Pulmonary embolism comes to mind immediately.

CONTEXT

Availability in the clinic

Residents encountered new cases with **similar findings but different diagnoses** after reviewing earlier cases.

1.55
Similar findings

2.19
Other cases

Diagnostic accuracy score / 4.

Mamede et al., JAMA 2010

CONTEXT

Can a number move your estimate?

GROUP A	GROUP B
10%	65%

What percentage of UN member states were African in 1974?



CONTEXT

Anchoring in a diagnostic workup

People tend to anchor to their initial starting points (e.g. chief complaints)

25%
Median UN estimate
Anchor: 10

45%
Median UN estimate
Anchor: 65

CONTEXT

Clinicians tend to stuck to first hypothesis

Referral Note

Prior Assessment

"Migraine"
An earlier hypothesis

→

New Findings

Current Visit

New focal neurological deficit and a sudden, severe headache.

CONTEXT

One correct diagnosis may still be incomplete.

Patient Presentation

Fever, cough, and a focal chest infiltrate.

Also: intense nocturnal itch and lesions between the fingers.

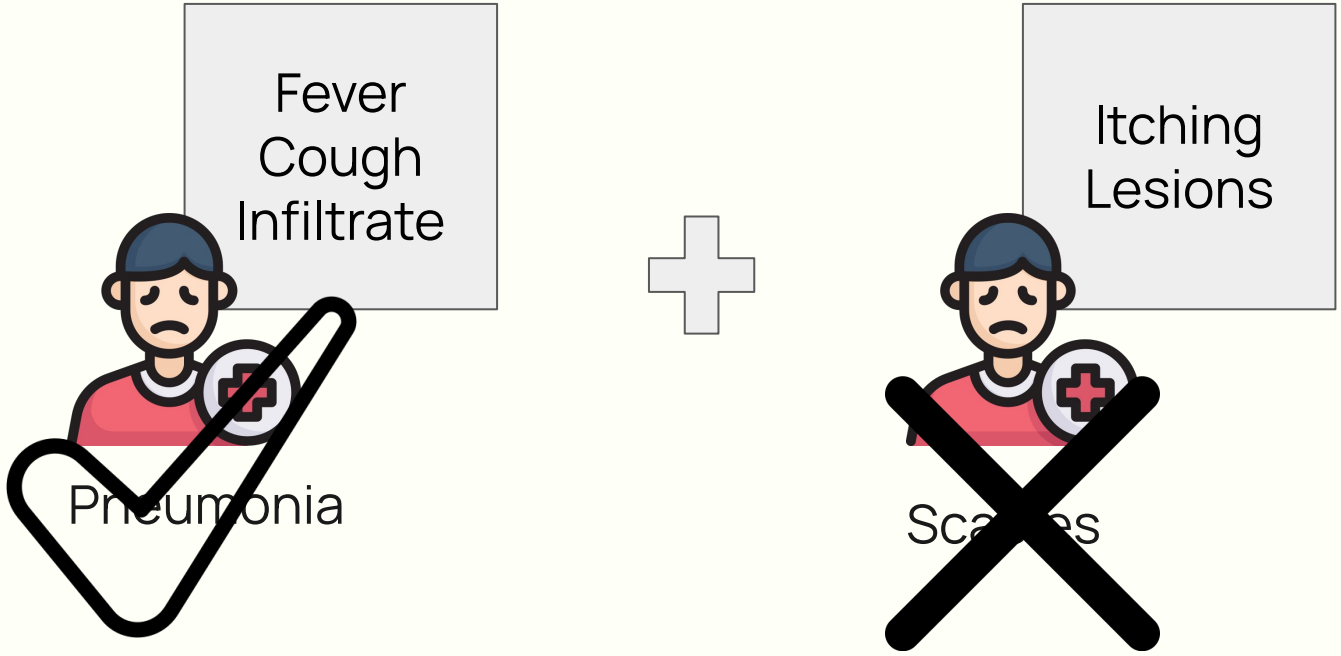
"Pneumonia explains the respiratory findings."

A Stop: the diagnosis is found

B Continue: an important finding remains

CONTEXT

You found one diagnosis. Are you finished?



CONTEXT

How much did confidence fall?

- **Easy cases:** 55.3% correct.
  Confidence: 7.2 / 10.
- **Hard cases:** 5.8% correct.

Confidence on the hard cases?

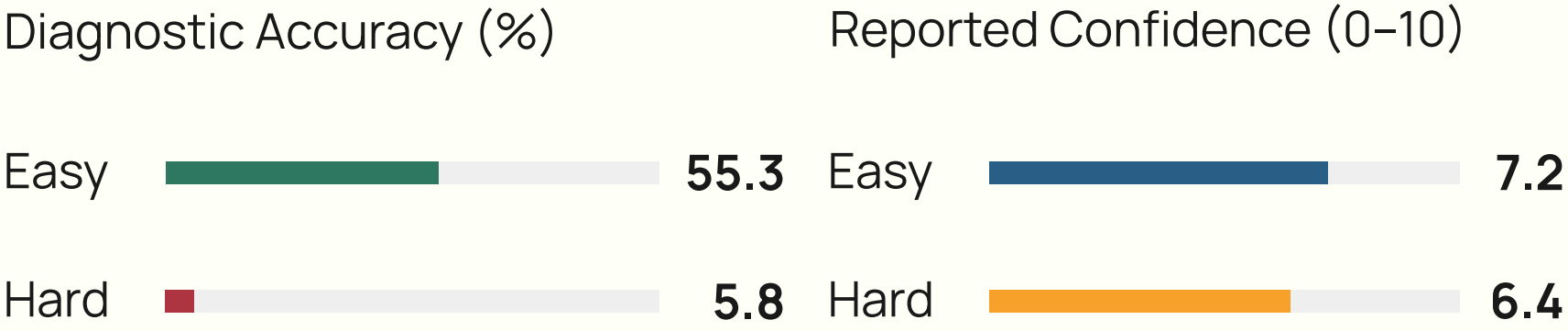
A ≈1 / 10 **B** ≈3 / 10 **C** ≈6 / 10



Meyer et al., 2013

CONTEXT

Overconfidence: confidence barely moved



Feeling **certain** is not the same as being correct.

Meyer et al., 2013

CONTEXT

At which handoff does “maybe” become “known”?

Could this be an eating disorder?



History of eating disorder



Known eating disorder



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CONTEXT

Repeated claims are not independent evidence.

Observed Evidence		Copies of the claim
1	≠	3
Original assessment		Confident records

CONTEXT

Other ways judgment can fail

- **Attribution** — Patient characteristics or labels distort the interpretation of clinical findings.
- **Base-rate neglect** — Ignoring prior probabilities when evaluating new evidence.
- **Confirmation Bias** — Favoring evidence that supports an existing hypothesis while overlooking contradictory findings.

CONTEXT

Cognitive and system factors in diagnostic error

Review of **100 diagnostic error cases**

74%
Cognitive contributors

65%
System contributors

The same case can involve **both** types of contributor.

CONTEXT

How to fix this?

We know clinicians are vulnerable to cognitive biases.

- Can we simply teach them to think more carefully?
- Is **analytical reasoning** the solution?
- Is **knowing about biases** enough?
- What actually helps?

CONTEXT

Can’t we just think more analytically?

204 assistant doctor solves 20 clinical case

44.5%
Think Fast

45.0%
Think Slow



Thinking longer does not
necessarily mean thinking better

Norman et al., 2014

CONTEXT

Can we teach clinicians to avoid bias?

191 medical students in a controlled trial

45%
Bias training

45%
Control group



Teaching cognitive forcing did not improve performance in this trial.

Sherbino et al., 2014

CONTEXT

What about structured reflection?

1.55

Initial non analytical reasoning

2.03

After structured reflection

Reflection checks **supporting evidence, conflicts and alternatives**.

Second-year residents, same cases. Score / 4.

Mamede et al., JAMA 2010

CONTEXT

What about better knowledge?

91 residents in a controlled trial

40%
Training on test set

24%
No training



Better knowledge can make reasoning more reliable.

Sherbino et al., 2014

CONTEXT

When clinical intuition deserves trust

01 **Stable patterns**

Clinical cues predict outcomes reliably.

02 **Repeated experience**

Similar cases offer opportunities to learn.

03 **Useful feedback**

Follow-up reveals correct and missed diagnoses.

Kahneman & Klein, 2009

CONTEXT

Back to Anne Dodge



- 15 years, nearly 30 doctors. Repeatedly diagnosed with anorexia, bulimia, and psychological problems. Down to 37 kg
- Her boyfriend insisted she see another specialist
- Dr. Myron Falchuk questioned previous assumptions, listened to her story from the beginning

Celiac Disease

Groopman, 2007

CONTEXT

Back to Anne Dodge

- **Anchoring** — Keeping the “eating disorder” label despite new evidence.
- **Confirmation bias** — Reading continued weight loss as support for the existing label.
- **Diagnostic momentum** — Treating a diagnosis passed between clinicians as an established fact.
- **Search Satisfaction** — Stopping the search once a plausible explanation is available.

Groopman, 2007

CONTEXT

What should a safer system do?

- Represent the patient's problem.
- Keep plausible diagnoses in play.
- Ask questions that distinguish them.
- Update with new evidence.
- Act or defer with uncertainty visible.
- Probabilistically grounded



Questions?

Building Systems to Make *Reliable* High Stakes Decisions

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Outline

01 Introduction

02 What is the problem?

03 Why does it happen?

04 What is there?

05 MoBayes

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Building Systems to Make Reliable High Stakes Decisions

- **System**
- **Reliable**
- **High Stakes**
- **Decision**

This is conversational clinical decision support.

Ledley & Lusted, 1959 · Shortliffe, 1976 · Miller et al., 1982

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Clinical decision support before LLMs

1959 — **Ledley & Lusted**
Probability and logic

1976 — **MYCIN**
Expert-written rules

1982 — **INTERNIST-I**
A curated disease–finding KB



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Medical Decision with Language Models

MedGemma



MEDITRON

Open medical training

Meditron-3

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Medical LLMs: strong exam performance

91.1% **Med-Gemini on MedQA**

What does an exam score leave **untested**?

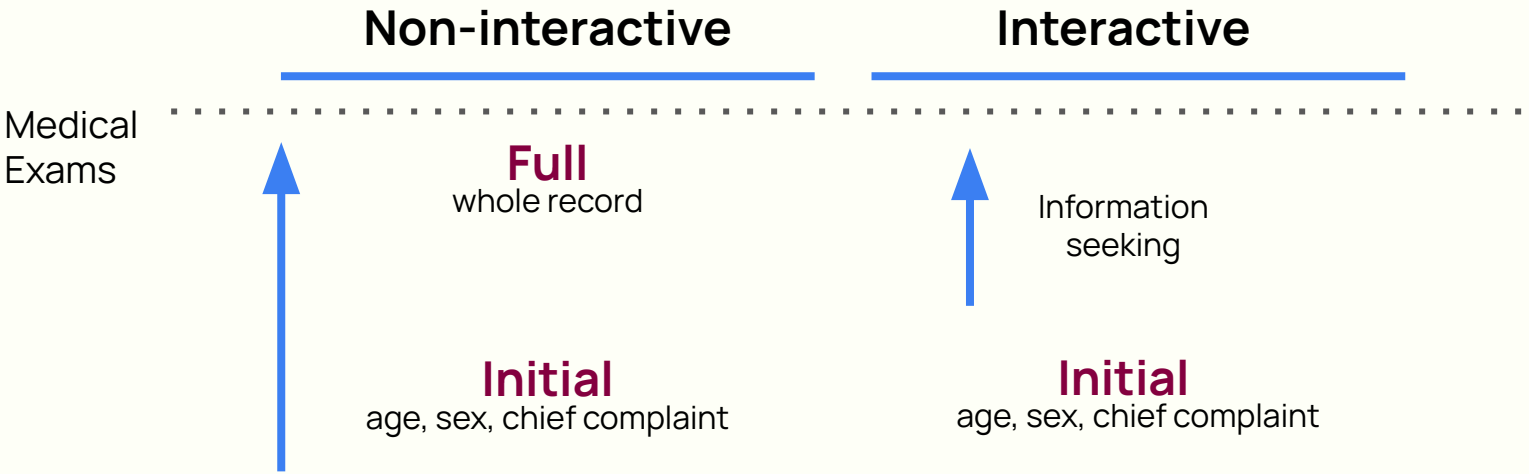
Why are they **not** used in **real practice**?

- Regulations
- Real life is noisy

High Stakes

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Clinical decision making is reasoning under uncertainty



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The target: a realistic clinical consultation

Incomplete patient answers

A limited question budget

Several possible conditions

Abstention when unresolved

(...)



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LLMs also include medical decision failures

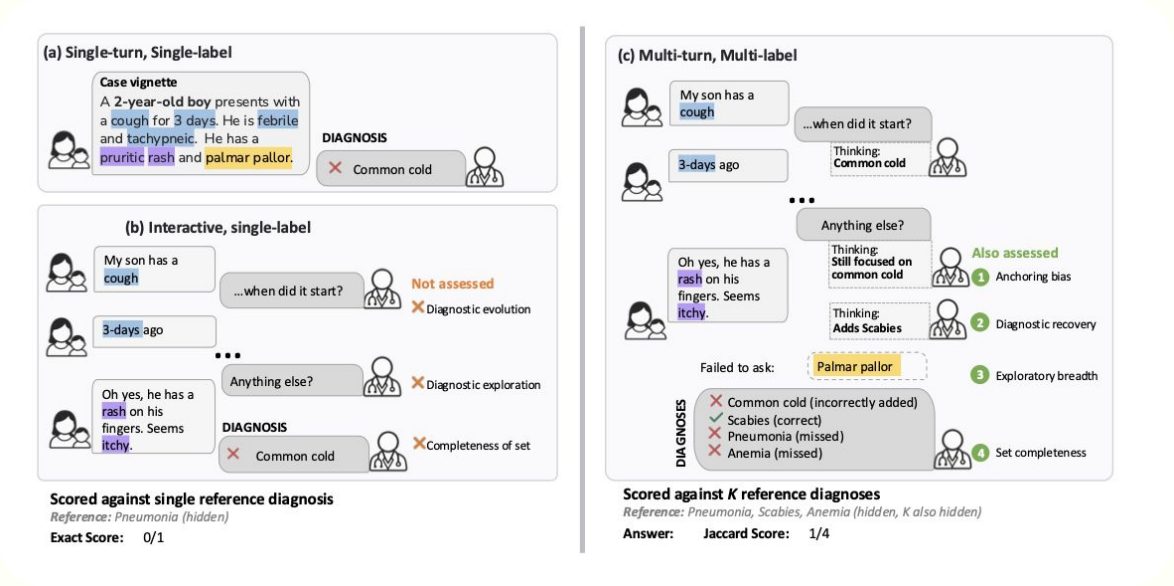
It is empirically shown that LLMs have problem with

- Anchors
- Search Satisfaction
- Framing
- Overconfidence
- etc

You can test it yourself.

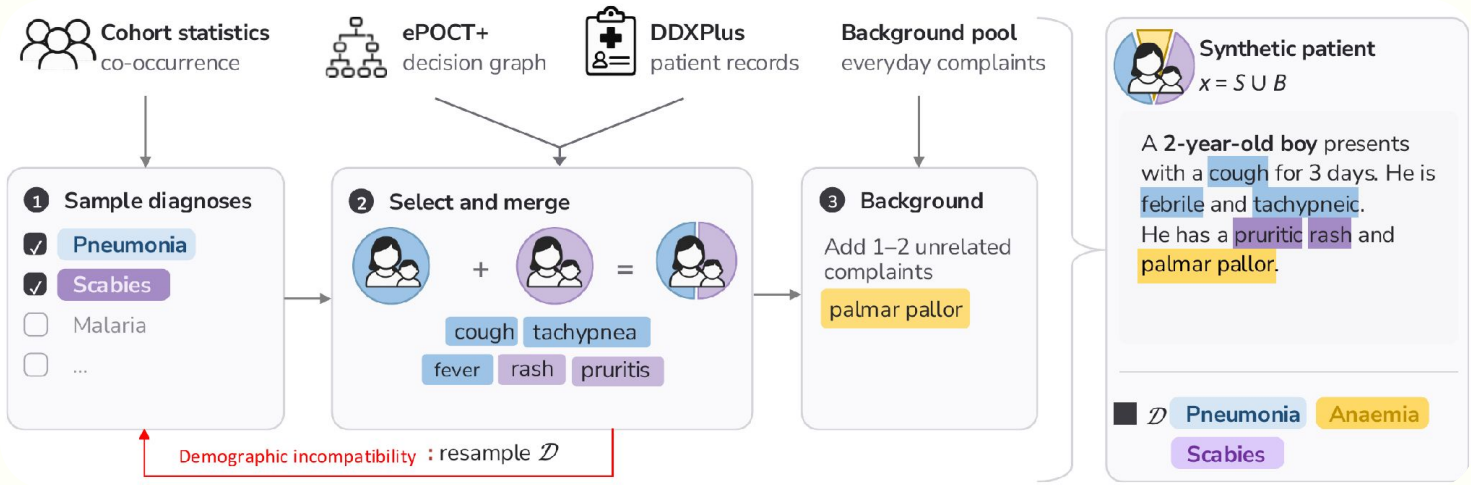
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An example: LLMs are single hypothesis trackers



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CLIMB: constructing a patient with several diagnoses



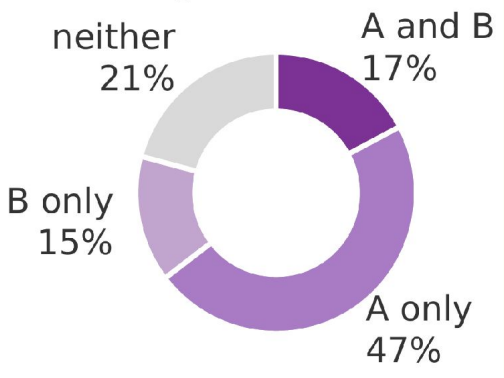
Recover the **complete diagnosis set**, with the true count hidden.

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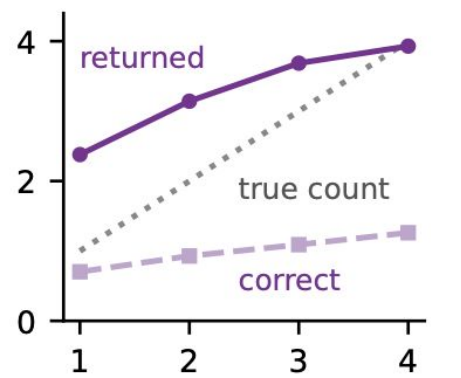
CONTEXT ENGINEERING

CLIMB: the opening diagnosis dominates

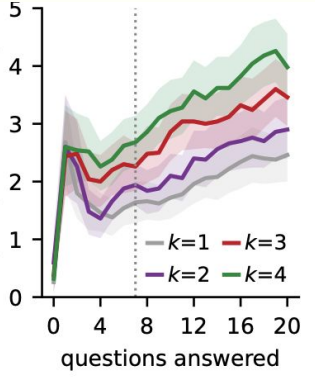
Models often retain the opening diagnosis. Asking for more diagnoses can add unsupported alternatives.



Opening diagnosis dominates



Correct count plateaus



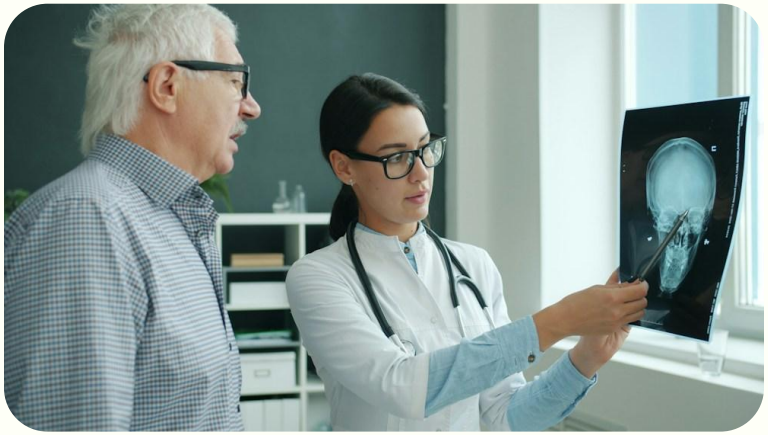
Wrong diagnoses accumulate

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CONTEXT ENGINEERING

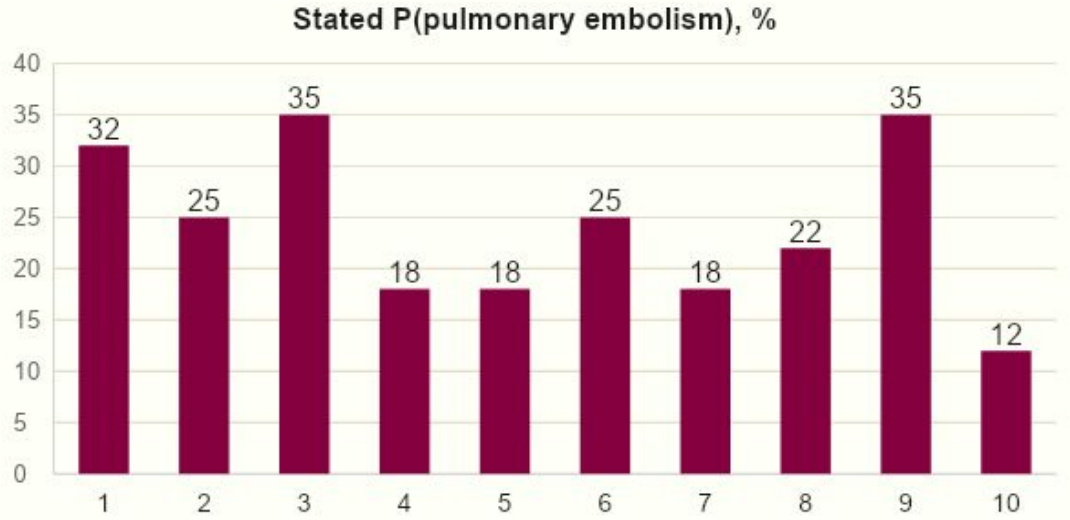
Why do these failures happen?

- Inconsistent probabilities
  Estimates keep changing
- Coarse estimates
  Subtle differences disappear
- Overconfidence
  Confidence exceeds accuracy
- Hallucination
  Unsupported claims appear



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For example: Probability is not a stable belief



12–35%

Same prompt and complaint.

It is also very coarse grained.

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Because: what does the training objective reward?

Pretraining

Predict the next token.

Learn data

$$\mathcal{L}(\theta) = -\sum_{t} \log P_\theta(x_t \mid x_{<t})$$

Post-training

Imitate demonstrations.

Learn preference

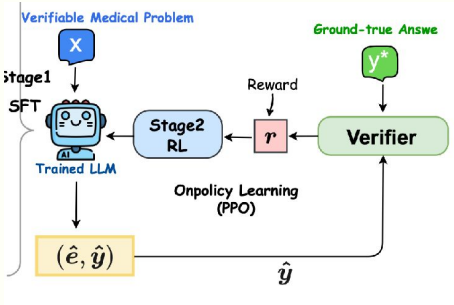
However, data and reward choices can carry **human biases**.

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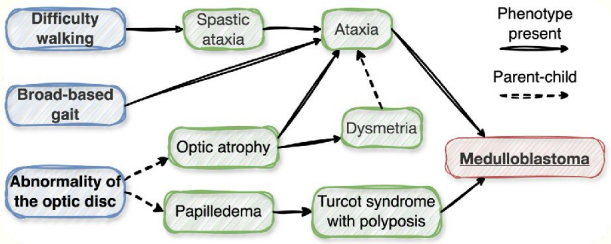
Reasoning models and "System 2"

- Consume some **more tokens** before giving the answer
- Isn't it like **system 2** thinking? We spend some time rather than giving answer directly
- But, reasoning models also have the **similar problems**.

HuatuoGPT-o1



MedReason-8B



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Structuring the Reasoning Process

- Chain of Thought: Explicit sequence of steps
  - Are the premises correct?
- Self-reflection: Critique and revision of an answer
  - Did it repeat the same assumptions?
- Self-consistency: Branch and do majority voting
- Fleet of Agents: Evolutionary search in solution space
- (...)

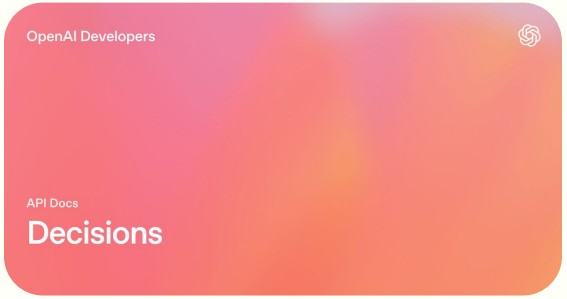
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Decision models

Decision models are gaining attention for fast choices among defined options. They return structured scores that software can act on.







Jev · TypeSafe

Microsoft-Decision-1

OpenAI Decisions API

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Diagnosis is abductive reasoning

- **Deduction** predicts findings from a hypothesis.
- **Induction** learns patterns from examples.
- **Abduction** compares explanations of observed findings.

We need proper probabilistic structures.

- BedLLM, BIRD, DeLLMa, ...

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BED-LLM: which question is informative?

Current history: $h_2 = ((\text{“Born in the 20th century?”}, \text{“Yes”}), (\text{“Is this person male?”}, \text{“Yes”}))$.

(A) **Extract beliefs (§3.1).** Construct hypothesis set $\Theta^{\text{cand}} = \{\theta_n\}_{n=1}^{N}$ by sampling candidate hypotheses, $\theta \sim p_{\text{LLM}}(\theta; h_2)$, and rejecting any θ that is inconsistent with h_2. Use joint model $p(\theta, y; h_2, x) = p_f(\theta; h_2) p_{\text{LLM}}(y; [\theta, x])$, where $p_f(\theta; h_2)$ is uniform over Θ^{cand}.

Barack Obama Steve Irwin Hugh Laurie Banksy Elvis Presley …

(B) **Generate candidate questions (§3.2).** Sample $\tilde{x}_{1:M} \sim p_{\text{LLM}}(\tilde{x}_{1:M}; [h_{t-1}, \Theta^{\text{cand}}])$.

$\tilde{x}_1$: “Was this person born in Antarctica?”
$\tilde{x}_2$: “Was this person born in the 19th century?”
$\tilde{x}_3$: “Is this person an artist?”
$\tilde{x}_4$: “Is this person European?”
$\tilde{x}_5$: “Does this person prefer thrash metal over death metal?”

(C) **Estimate EIG (§3.3).** Compute $\underbrace{\mathrm{H}[\hat{p}(y; [h_2, x])]}_{\text{marginal entropy}} - \underbrace{\frac{1}{N}\sum_{n=1}^{N} \mathrm{H}[p_{\text{LLM}}(y; [\theta_n, x])]}_{\text{expected conditional entropy}} \approx \mathrm{EIG}_\theta(x; h_2)$.

	Question	Marg.	Exp. cond.	EIG	Intuition
$\tilde{x}_1$	Born in Antarctica?	≈ 0	≈ 0	≈ 0	Answer “No” for all θ_n
$\tilde{x}_2$	Born in 19th century?	≈ 0	≈ 0	≈ 0	Redundant given h_2
$\tilde{x}_3$	An artist?	0.56	0.41	0.15	Uneven split
$\tilde{x}_4$	European?	0.97	0.17	0.80	Balanced split; crisp answers
$\tilde{x}_5$	Thrash vs. death metal?	0.89	0.88	0.01	Uncertain even given θ

(D) **Select and ask the best question.** Choose $x_3 = \tilde{x}_4$ by maximizing EIG. Output this to the user.

(E) **Update the history.** Observe answer, y_3. Set $h_3 = (h_2, (x_3, y_3))$.

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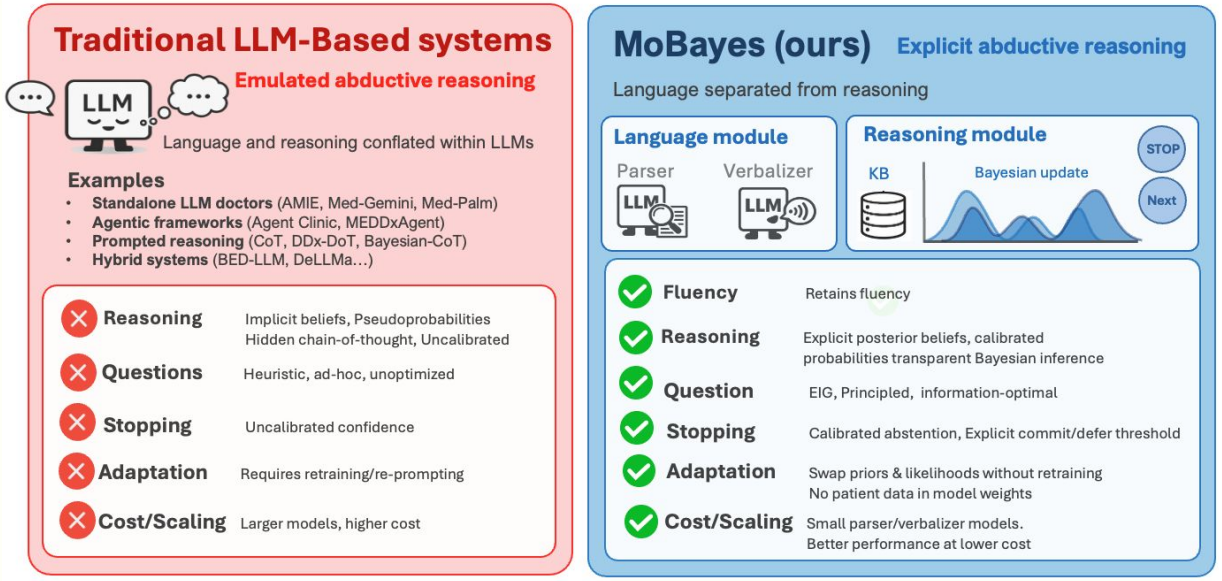
Clinical reasoning needs

- A **belief distribution** over diseases
- A numerical objective for the next question
- A policy for **answering or deferring**
- An output that represents uncertainty



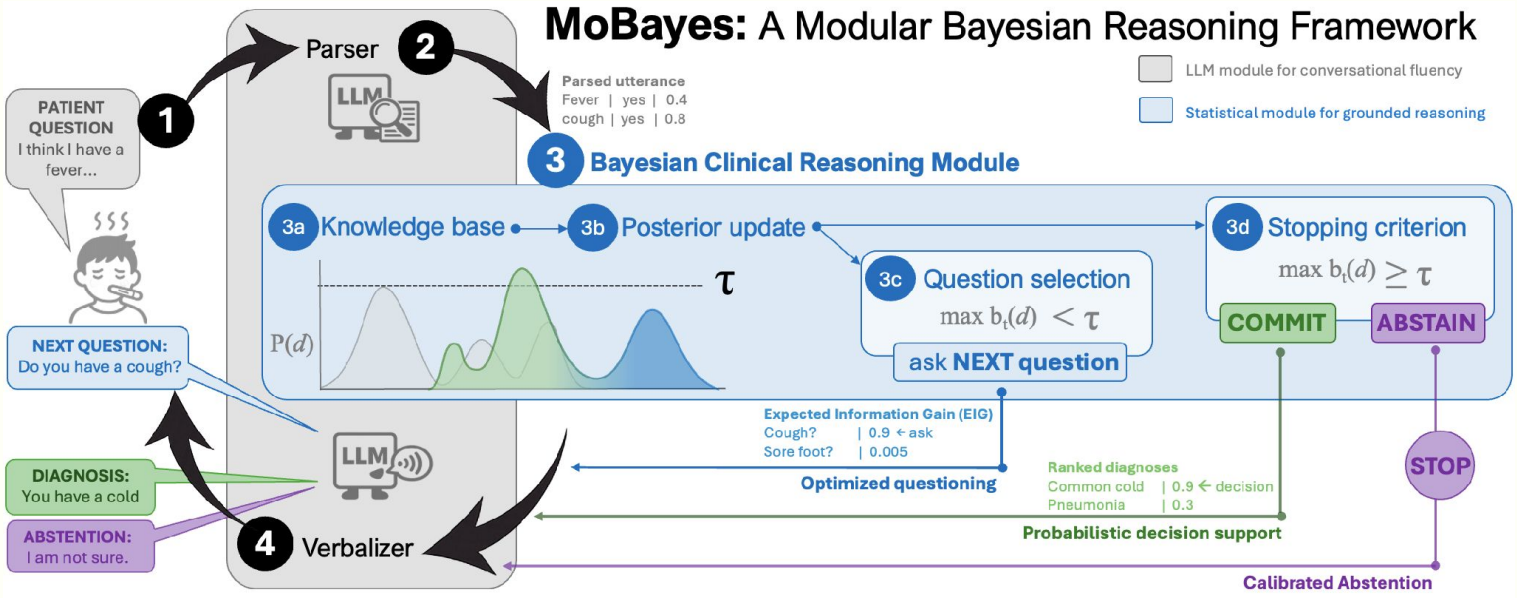
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MoBayes separates language from inference



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MoBayes: the full consultation loop



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Building a knowledge base from records

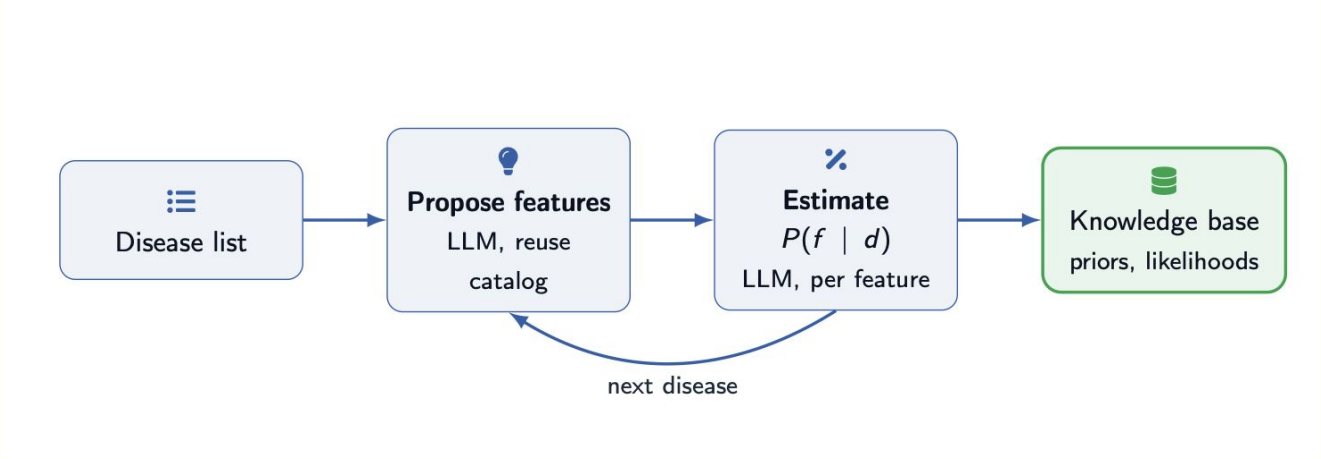
Hospital records should be used:

- To generate a probability table
- It defines a closed universe. When evidence is insufficient, refer to clinician

Disease	Fever	Cough
Pneumonia	0.80	0.85
Bronchitis	0.35	0.90
Asthma	0.10	0.40

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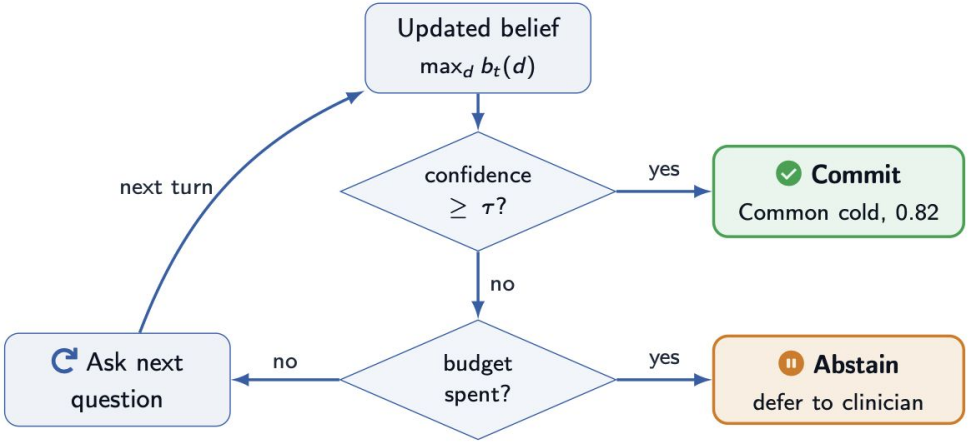
Building an external KB from an LLM



Elicit probabilities once, then **inspect and validate** the table.

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Abstention is a decision policy



A warm-up phase blocks early commits; τ tunes triage versus safety.

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Belief update and question selection

BELIEF	Track the current disease probabilities.
UPDATE	Use new evidence to revise those probabilities.
NEXT QUESTION	Ask what is expected to reduce uncertainty most.

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Uncertain evidence has less influence

"I think I lost weight."

Parser confidence: $0 \leq c \leq 1$

$$\text{Adjusted likelihood} = c \times P(v \mid d) + (1 - c)$$

$c = 1$ Usual likelihood update

$c = 0$ Belief stays unchanged

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Accuracy and coverage must be scored together

We need new metrics to define the trade-off between selective accuracy. We want AI doctors that are never wrong!

90%

Selective accuracy: 54 / 60

60%

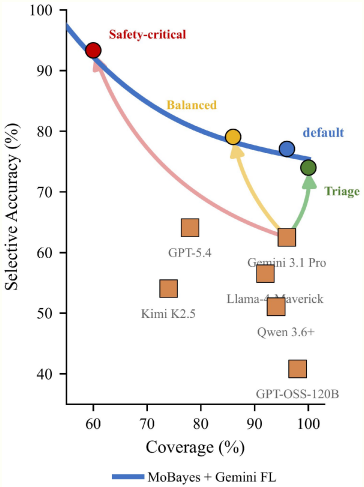
Coverage: 60 / 100

$$\text{DHS} = \frac{2AC}{A + C} = \frac{2 \times 0.90 \times 0.60}{0.90 + 0.60} = 0.72$$

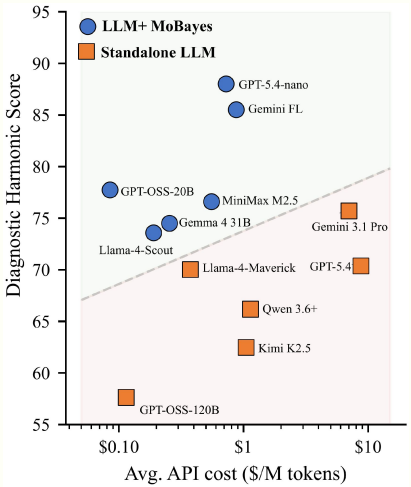
ENGINEERING

MoBayes: three results to examine

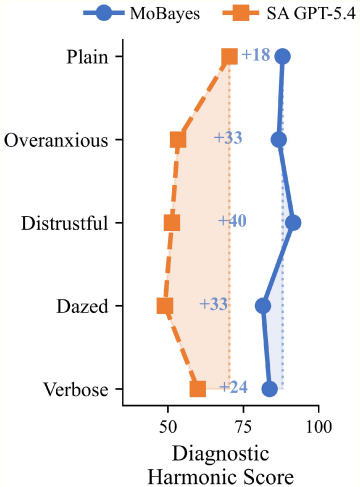
Controllable



Cost Efficient



Robust



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ENGINEERING

CLINIVERSE: scaling the knowledge base

- **Extract** medical knowledge from LLMs.
- **Build** a structured knowledge base.
- **Make** it inspectable and reusable.



Questions?

Building intelligent Global Health and Humanitarian Response Technologies

Thank You

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